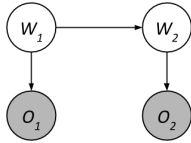


1 Particle Filtering

Let's use Particle Filtering to estimate the distribution of $P(W_2|O_1 = a, O_2 = b)$. Here's the HMM again. O_1 and O_2 are supposed to be shaded.



W_1	$P(W_1)$
0	0.3
1	0.7

W_t	W_{t+1}	$P(W_{t+1} W_t)$
0	0	0.4
0	1	0.6
1	0	0.8
1	1	0.2

W_t	O_t	$P(O_t W_t)$
0	a	0.9
0	b	0.1
1	a	0.5
1	b	0.5

We start with two particles representing our distribution for W_1 .

$P_1 : W_1 = 0$

$P_2 : W_1 = 1$

Use the following random numbers to run particle filtering:

[0.22, 0.05, 0.33, 0.20, 0.84, 0.54, 0.79, 0.66, 0.14, 0.96]

(a) **Observe:** Compute the weight of the two particles after evidence $O_1 = a$.

$$w(P_1) = P(O_t = a|W_t = 0) = 0.9$$

$$w(P_2) = P(O_t = a|W_t = 1) = 0.5$$

(b) **Resample:** Using the random numbers, resample P_1 and P_2 based on the weights.

We now sample from the weighted distribution we found above. Using the first two random samples, we find:

$$P_1 = \text{sample}(\text{weights}, 0.22) = 0$$

$$P_2 = \text{sample}(\text{weights}, 0.05) = 0$$

(c) **Predict:** Sample P_1 and P_2 from applying the time update.

$$P_1 = \text{sample}(P(W_{t+1}|W_t = 0), 0.33) = 0$$

$$P_2 = \text{sample}(P(W_{t+1}|W_t = 0), 0.20) = 0$$

(d) **Update:** Compute the weight of the two particles after evidence $O_2 = b$.

$$w(P_1) = P(O_t = b|W_t = 0) = 0.1$$

$$w(P_2) = P(O_t = b|W_t = 0) = 0.1$$

(e) **Resample:** Using the random numbers, resample P_1 and P_2 based on the weights.

Because both of our particles have $X = 0$, resampling will still leave us with two particles with $X = 0$.

$$P_1 = 0$$

$$P_2 = 0$$

(f) What is our estimated distribution for $P(W_2|O_1 = a, O_2 = b)$?

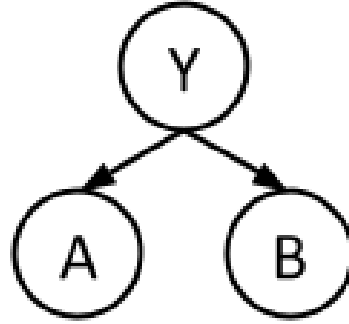
$$P(W_2 = 0|O_1 = a, O_2 = b) = 2/2 = 1$$

$$P(W_2 = 1|O_1 = a, O_2 = b) = 0/2 = 0$$

2 Naive Bayes

In this question, we will train a Naive Bayes classifier to predict class labels Y as a function of input features A and B . Y , A , and B are all binary variables, with domains 0 and 1. We are given 10 training points from which we will estimate our distribution.

A	1	1	1	1	0	1	0	1	1	1
B	1	0	0	1	1	1	1	0	1	1
Y	1	1	0	0	0	1	1	0	0	0



1. What are the maximum likelihood estimates for the tables $P(Y)$, $P(A|Y)$, and $P(B|Y)$?

Y	$P(Y)$	A	Y	$P(A Y)$	B	Y	$P(B Y)$
0	$3/5$	0	0	$1/6$	0	0	$1/3$
1	$2/5$	1	0	$5/6$	1	0	$2/3$
		0	1	$1/4$	0	1	$1/4$
		1	1	$3/4$	1	1	$3/4$

2. Consider a new data point ($A = 1, B = 1$). What label would this classifier assign to this sample?

$$P(Y = 0, A = 1, B = 1) = P(Y = 0)P(A = 1|Y = 0)P(B = 1|Y = 0) \quad (1)$$

$$= (3/5)(5/6)(2/3) \quad (2)$$

$$= 1/3 \quad (3)$$

$$P(Y = 1, A = 1, B = 1) = P(Y = 1)P(A = 1|Y = 1)P(B = 1|Y = 1) \quad (4)$$

$$= (2/5)(3/4)(3/4) \quad (5)$$

$$= 9/40 \quad (6)$$

$$(7)$$

Our classifier will predict label 0.

3. Let's use Laplace Smoothing to smooth out our distribution. Compute the new distribution for $P(A|Y)$ given Laplace Smoothing with $k = 2$.

A	Y	$P(A Y)$
0	0	$3/10$
1	0	$7/10$
0	1	$3/8$
1	1	$5/8$